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Letter Section

Tectonic implications of a new palaeomagnetic result from the Jurassic of the Transdanubian Central Mountains

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ABSTRACT

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The analysis of the natural remanent magnetization (N.R.M.) of the Jurassic limestones from the Transdanubian Central Mountains (T.C.M's) (6 sites, 137 samples) resulted in the following palaeomagnetic pole position: $\Phi = 54°$, $\Lambda = 289°$ with $\alpha_{95} = 11°$.

This result is explicable from two entirely different points of view as follows:

(1) The T.C.M's may always have formed part of stable Europe but rotated 73° counterclockwise about a vertical axis relative to it.

(2) The T.C.M's may have formed part of the African plate in the Adriatic region, shared the movements of Africa relative to stable Europe until when rotated to some degree counterclockwise about a vertical axis.

INTRODUCTION

Three Jurassic type localities of the Transdanubian Central Mountains (T.C.M's) (Bakonycsernye, Lókut, Tata-Kálváriadomb) were selected for palaeomagnetic studies. The fourth locality (Tardosbánya) was sampled for greater areal coverage.

The natural remanent magnetization (N.R.M.) of all samples was treated by AF-demagnetization using a Schonstedt GSD-1 demagnetizer. Several pilot specimens were demagnetized stepwise up to a peak field of 300—1000 Oe depending on the hardness of the magnetization and the sensitivity of our magnetometers (JR-2 and JR-4 spinners). From the remainder, one specimen of each sample was demagnetized in a single step at the optimum cleaning field the latter having been judged from the orthogonal demagnetization graphs of the pilot specimens (Fig. 1). Figure 1 shows a

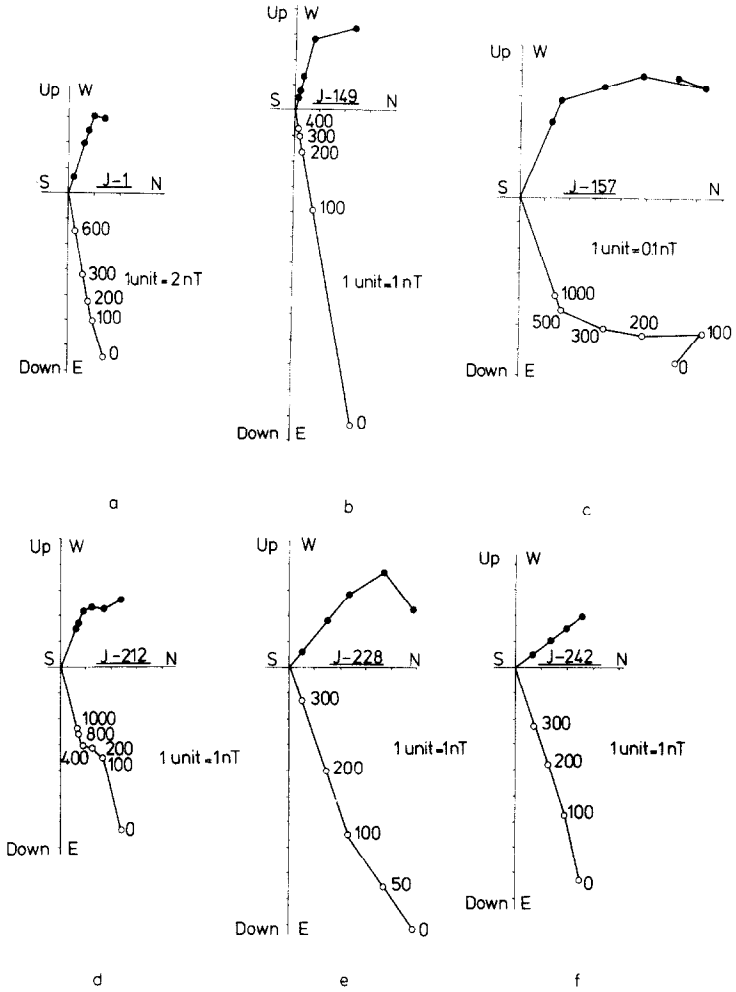


Fig.1. A selection of orthogonal AF-demagnetization graphs, one for each sampling site. a. Bakonycsernye section A (Lower Jurassic). b. Bakonycsernye section A (Middle and Upper Jurassic). c. Bakonycsernye section B. d. Lókut. e. Tata-Kálváriadomb. f. Tardosbánya. End points of magnetic vectors are represented by open symbols on the N—S vertical plane and by full symbols on the horizontal plane. Numbers beside plotted points denote peak demagnetization field in Oersteds.

variable hardness of the magnetizations but the stability of the N.R.M. is indicated by the consistency of the site mean directions and their being significantly different from the present field direction.

All relevant palaeomagnetic data presented for the purpose of this letter are summarized in Table I. Details of the palaeomagnetic studies will be published elsewhere.

TABLE I

Summary of palaeomagnetic results from the Jurassic of the Transdanubian Central Mountains

Sampling site	Geographic coordinates		No. of samples used	Demagnetizing field (Oe)	Site mean direction of N.M.R. corrected for DIP of strata		Statistical parameters		Palaeomagnetic pole position		Statistical parameters	
	φ°	λ°			$\bar{D}^\circ$	$\bar{I}^\circ$	K	α°_{95}	Φ°	Λ°	Δp	Δm
Bakonycsernye section A (Lower Jurassic)	47.29	18.11	69	300	306.7	55.6	41.8	2.7	49.5	292.2	2.7	3.8
Bakonycsernye, section A (Middle and Upper Jurassic)	47.29	18.11	16	300	325.9	58.2	45.9	5.5	64.0	281.9	6.0	8.1
Bakonycsernye, section B (Lower Jurassic)	47.29	18.11	13	300	310.4	45.9	17.3	10.3	46.7	278.6	8.4	13.1
Lókut	47.20	17.88	23	500	293.6	44.9	15.2	8.0	34.8	290.6	6.4	10.1
Tata-Kálváriadomb	47.64	18.32	9	200	326.2	62.4	44.1	7.8	66.25	291.8	9.5	12.2
Tardosbánya	47.67	18.48	7	200	316.4	63.0	84.6	6.9	60.0	298.7	8.5	10.8
Mean	47.5	18.0							54	289	$\alpha_{95} = 11^\circ$	

DISCUSSION

The Jurassic poles for the T.C.M's and stable Europe (Heller, 1977) are widely different, but a 73° counterclockwise rotation about a vertical axis of the T.C.M's will send the palaeomagnetic pole precisely in to the stable European one (Fig. 2). This means that the present position of the T.C.M's fits very well in the system of palaeolatitudes for stable Europe in the Jurassic, but a large rotation of the region is required to restore its primary orientation. This movement is similar to those usually suggested for other regions in the Mediterranean area (Zijderveld et al., 1970; Lowrie and Alvarez, 1975).

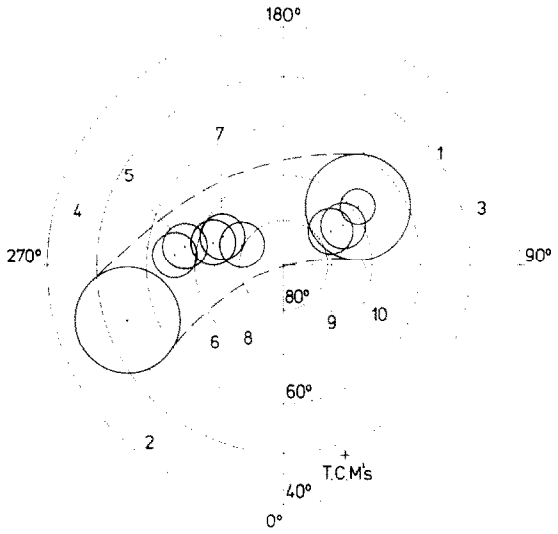


Fig.2. Polar stereographic projection showing Jurassic palaeomagnetic pole positions with 95% confidence circle (Fisher, 1953) with respect to present-day Europe. The upper Jurassic pole for stable Europe (*1*) is $\Phi = 69°$, $\Lambda = 128°$, $\alpha_{95} = 4°$ (Heller, 1977). The Jurassic pole for the Transdanubian Central Mountains (*2*), $\Phi = 54°$, $\Lambda = 289°$, $\alpha_{95} = 11°$, becomes (*3*): $\Phi = 69°$, $\Lambda = 129°$ when rotated 73° clockwise about a vertical axis in the sampling area ($\varphi = 47.5°$, $\lambda = 18°$). The Jurassic pole for Africa (*4*), $\Phi = 66°$, $\Lambda = 264°$, $\alpha_{95} = 5°$ was computed from nine poles in McElhinny (1973), McElhinny and Cowley (1977) and Márton and Márton (1976). Successive plotted positions of this pole due to the movements of Africa relative to stable Europe, according to Dewey et al. (1973), are as follows:

5:	$\Phi = 68°$	$\Lambda = 259°$	9 m.y. ago.
6:	74°	252°	53
7:	75°	245°	63
8:	80°	244°	80
9:	77°	125°	148
10:	74°	123°	180

All figures given are rounded to the next integer.

Contrary to the autochtonous position of the T.C.M's implied by the above interpretation of our palaeomagnetic result there seems to exist geologic (palaeontologic and facial) evidence that the Jurassic of the T.C.M's originates from the southern margin of the Tethys as part of the now dissected large carbonate belt which stretches through the Mediterranean (Laubscher, 1971; Géczi, 1973; Bernoulli and Jenkins, 1974).

The palaeolatitude of the T.C.M's is 37 ± 11° and if their Jurassic limestones deposited somewhere in the southern Tethys it could only be in the Adriatic region which may then have formed part of the African plate (see e.g., Channel and Horváth, 1976).

The supposition that the Jurassic limestones of the T.C.M's formed in the Adriatic region and that the latter was attached to the African plate is supported by the closeness of the Jurassic palaeomagnetic poles for the T.C.M's and Africa (Fig. 2). In fact, the palaeolatitude computed from the African pole is 32 ± 5°, practically equal to that determined from the measured inclination (37 ± 11°).

The existing significance of the angular distance between the Jurassic poles of the T.C.M's and Africa arises from a small counterclockwise rotation of the declination of the T.C.M's result with respect to that computed from the African pole. Consequently, all that is required by our palaeomagnetic data to make the T.C.M's pole "African" is that this declination difference be removed simplest by untwisting the region about a vertical axis.

In Fig. 2 we included also the successive positions of the Jurassic palaeomagnetic pole (with confidence circle) for Africa relative to present-day Europe for the end of each interval in which the movement of Africa with respect to Europe may be approximated by a finite rotation according to Dewey et al. (1973).

If the T.C.M's had been on the Adriatic (African) plate then the Jurassic poles of the T.C.M's and Africa should have overlapped until the Adriatic region separated from Africa. Figure 2 shows that, indeed, the T.C.M's pole can be brought to an insignificant distance to any one of the moved Jurassic poles for Africa by simply rotating the T.C.M's about a vertical axis and so doing we will have the whole track of the African pole fully contained in an 11° wide belt which covers the possible positions of the Jurassic palaeomagnetic pole of the rotated T.C.M's. Thus, the present palaeomagnetic data do not indicate which of the moved Jurassic poles of Africa may have been the last "African" pole for the T.C.M's.

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